CPS610-W26, Assignment-3, Labs 4, 5 – Database Systems-II **Due: Mar5, 2026, 4pm**

(The TA will evaluate your work and ask questions about progress and results in the Labs: weeks of Feb23 & Mar2)

Objectives

- Part1: Understanding and working with Transaction Management System
- Part2: Simulating Serializable Schedules for Transactions and Detecting Conflicts
- Part3: Testing Serializability for Transaction Schedules using Precedence Graphs

Part1- Serializability of Schedule of Transactions [16 Marks]

Most concurrency control methods on Transactions do not actually test for Serializability. Rather protocols and rules are developed to guarantee that any Schedule that follows those rules will be Serializable. To gain a better understanding of Concurrency Control Protocol on Transactions, specific Algorithm was discussed in class for testing Conflict-Serializability through transactions. In this assignment the relevant Algorithm will be simulated following some practice on testing the Serializability on selected transactions.
For this part, consider the three transactions T1, T2 and T3 in figure-1, and answer the following questions:

Transaction T_1
read_item(X);
write_item(X);
read_item(Y);
write_item(Y);

Transaction T_2
read_item(Z);
read_item(Y);
write_item(Y);
read_item(X);
write_item(X);

Transaction T_3
read_item(Y);
read_item(Z);
write_item(Y);
write_item(Z);

Figure-1

a) How many possible schedules exist for these three transactions? Describe your answer showing the total number and details about the transactions' orders (with operations). Are the schedules all Serialized or not (discuss briefly)?

b) Write a program to create all possible schedules for the three transactions in Figure-1.

c) Write a program to simulate the Algorithm for "Testing Conflict-Serializability" discussed in class, for each input schedule S.

d) Using your program in part (c) for the inputs from Part-2 (following section), determine which of those schedules are Conflict-Serializable and which are not.
For each Conflict-Serializable schedule, your program should print/display that schedule and list all equivalent serial schedules.
For each non-Serializable schedule, your program should print/display that schedule and the exact issues with it.

Note - Possible schedules should be based on the whole Transactions or the Operations inside each Transaction. For simplicity in your work consider the option of 1-3 Transactions, but your program should be able to detect the serializability over any list of schedules and operations.
Note - Preferably use an object-oriented language in your programming.

Part2- Precedence Graphs [8 Marks]

Consider the following schedules, where R, W, C stands for 'Read', 'Write' and 'Commit' respectively and subscripts refers to transactions (i.e., R1(X) means transaction T1 read X):

Schedule 1: R1(X), W1(X), R3(X), W3(X), W2(X), R1(Y), W1(Y), C1, W3(Y), C3, R2(Y), W2(Y), C2

Schedule 2: R2(X), W2(X), R3(Y), W3(Y), R3(Z), W3(Z), C3, R2(Z), W2(Z), C2, R1(X), W1(X), C1

Schedule 3: W1(A), W2(B), W3(C), R1(X), R2(X), R1(Y), W1(X), C1, W2(Y), C2, W3(Y), C3

For each schedule answer the following question:

(a) Draw the Precedence Graph for each schedule, and then determine if the schedule is Serializable or not?
(b) If the schedule is Serializable, can you provide other possible equivalent Serializable Schedules (at least 2 cases)?
(c) If not, discuss about the issues found (at least 2 cases). Then determine possible re-ordering to make those schedules (2 cases) to be Serialized?

Deliverables & Marking Schema

- Students are required to go through the instructions above, discuss their solution via team work, and finally submit their solution through D2L by the due date. Only ONE submission is required for each team.
- Teams are also required to indicate their partial work to the TA, asking their questions and getting directions from the TA during the lab times until the due.
- The TA will mark assignments by individual assessment (during the labs) for partial and then completed work as well as evaluation of the final submission.
  Lab attendance is mandatory to get the marks for both individual assessment and submitted document evaluations.
- The marks for Labs-4&5 (Assignment-3) will be based on the followings:
  1) Individual assessments in the Lab sessions during the week of Feb23 and Mar2 (only for attended members)
  2) Evaluation of the submission by Mar5, 4pm (only for attended members in the Labs)
  [Total 24 Marks = weight 4%]